

Exercises on Hypothesis Testing

1. The company XYZ manufacturing bulbs hypothesizes that the life of its bulbs is 145 hours with a known standard deviation of 210 hours. A random sample of 25 bulbs gave a mean life of 130 hours. Using a 0.05 level of significance, can the company conclude that the mean life of bulbs is less than the 145 hours?

2. The manager of a hotel is trying to decide which of the two supposedly equally good cigarette-vending machines to install, tests each machine 500 times, and finds that machine-1 fails to work 26 times and machine-2 fails to work 12 times. Using a 0.05 level of significance, can he conclude that two machines are not equally good?

3. If in certain test, 54 boys qualified out of a random sample of 150, while 31 girls qualified out of random sample of 100, can we conclude at the 0.05 level of significance that there is no difference in the performance of boys and girls?

Testing of Hypotheses

4. Advertisements claim the average nicotine content of a certain kind of cigarette is 0.30 mg. Suspecting that this figure is too low, a consumer protection service takes a random sample of 15 of these cigarettes from different production lots and finds that their nicotine content has a mean of 0.33 mg with a standard deviation of 0.018 mg. Use the 0.05 level of significance to test the null hypothesis $\mu = 0.30$ against the alternative hypothesis $\mu > 0.30$.

5. In a study of the effectiveness of physical exercise in weight reduction, a group of 11 persons engaged in a prescribed programme of physical exercise for 45 days showed the following results:

S.No.	Weight before (pounds)	Weight after (pounds)	S.No.	Weight before (Pounds)	Weight after (Pounds)
1	209	196	7	158	159
2	178	171	8	180	180
3	169	170	9	170	164
4	212	207	10	153	152
5	180	177	11	183	179
6	192	190			

Use the 0.05 level of significance to test the null hypothesis that the prescribed programme of exercise is not effective in reducing weight.

6. In a departmental store's study designed to test whether the mean balance outstanding on 30-day charge account is same in its two suburban branch stores, random samples yielded the following results:

$$n_1 = 60 \quad X_1 = 6420 \quad s_1 = 1600$$

$$n_2 = 100 \quad X_2 = 7141 \quad s_2 = 2213$$

where the subscripts denote branch store 1 and branch store 2. Use the 0.05 level of significance to test the hypothesis against a suitable alternative.

7. A product is produced in two ways. A pilot test on 6th times from each method indicates that product of method 1 has sample mean tensile strength 53 kg and a standard deviation 6kg, whereas in method 2 the corresponding values of mean and standard deviation are 50kg and 5kg respectively. Greater tensile strength in the product is preferable. Use an appropriate large sample test of 5 per cent level of significance to test whether or not method 1 is better for processing the product. State clearly the null hypothesis.

8. Two types of new cars produced in India are tested for petrol mileage. One of group consisting of 36 cars averaged 14 km per litre while the other group consisting of 72 cars averaged 12.5 km per litre.

(a) What test statistic is appropriate if $\sigma_1 = 1.5$ & $\sigma_2 = 2.0$?

(b) Test, whether there exists a significant difference in the petrol consumption of two types of cars (use $\alpha = 0.01$).